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EDUCATION



Master of Science in Robotics Engineering | Worcester Polytechnic Institute (WPI), Worcester, USA | May 2021| GPA: 3.86/4



Bachelor of Technology in Electronics and Communication Engineering | Rajiv Gandhi Institute of Technology, Mumbai, India | June 2016|GPA: 8.64/10

CORE COMPETENCIES

Machine Vision Systems

Image Processing

Computer Vision

Algorithm Development

Automated Inspection

Data Analysis

Optical Character

Recognition/OCR

Robotic Vision

Application Development

Excellent Communications skills

Convolutional Neural Networks

GenAI with AWS

TECHNICAL SKILLS

Programming Language: C++,

Python, Matlab

Machine Learning: Tensorflow,

Keras, Pytorch, Numpy, Pandas,

Scikit-learn, OpenCV, PCL, GCP

Robotics : ROS, Gazebo, Rviz

Tools and OS : Linux(Ubuntu),

Windows, Git, GitHub,

VisualStudio, Jupyter, Anaconda,

Matplotlib,Data Modeling

PROJECTS

Driveable Area Detection with Attention UNET /Tools: Pytorch, Python,OpenCV, SkLearn

Developed and implemented an Attention-based UNet in PyTorch for drivable area detection, accurately identifying both main and adjacent lanes for driving using the Berkeley Dataset. Trained the model with varying parameters, achieving a Dice coefficient accuracy of 0.7. (Tools: PyTorch, Python, OpenCV, SkLearn).

Image Classification / Tools Python, Pytorch,CUDA

Implemented vision transformers for image classification on COCO dataset with 95% accuracy (Tools: Python, PyTorch, CUDA).

Behavioral Cloning for Autonomous Driving | Tools: CARLA,Pytorch,OpenCV,Python,Linux

Executed behavioral cloning for autonomousdriving by integrating the CARLA Simulator with Linux tocollect 2D images by camera mounted on head of car while driving in the simulation. Trained an encoder-decoder CNNmodel in PyTorch to estimate steering angles, reaching 90% accuracy. (Tools: CARLA, PyTorch, OpenCV, Python, Linux).

Semantic Segmentation / Tools Python, Pytorch, CUDA

Developed a novel convolution-based skip connection,improving upon traditional skip connections & achieving 98% accuracy on COCO dataset. (Tools: Python, PyTorch, CUDA).

SensorFusionof Camera and Lidar for3Dobject Tracking/Tools: OpenCV, C++, PCL

Performed sensor fusion for 3D objecttracking by detecting vehicles using YOLO-trained models andcalculating bounding boxes around cars. Clustered 3D pointcloud data with the RANSAC algorithm to associate LIDARpoints with vehicles in road scenes. (Tools: OpenCV, C++, PCL).

AKSHATA PORE

Computer Vision Engineer

JOB OBJECTIVE

Pursuing a senior role as a Machine Vision Engineer, leveraging over 4 years of hands-on experience in developing and deploying advanced **machine vision systems**. Aiming to apply expertise in **image processing**, **3D vision**, **AI integration**, and real-time **system optimization**

- PROFILE SUMMARY
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Developed a profound understanding of various roles throughout the career, including **software development**, **algorithm optimization**, and **project management**, while acquiring a diverse skill set in **machine learning and robotics**.
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Achieved a remarkable **95% detection** accuracy for Gear Case bungs through the innovative application of the **RCNN model**, meaningfully enhancing the operational capabilities of the **CSCV machine vision system**.
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Deep experience in **Camera Calibration, Configuration, and Optimization**, ensuring high-accuracy image capture and robust system performance.
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Possesses extensive knowledge of **advanced machine learning frameworks and tools**, including TensorFlow, Keras, and OpenCV, enabling the development of sophisticated perception algorithms.
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Recently **optimized the performance** of a critical subprocess, resulting in a **30% reduction** in processing time, thereby enhancing the overall efficiency of **robotic operations**.
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Exhibits a strong command of **robotics frameworks** such as **ROS and Gazebo**, facilitating the development of **autonomous systems** capable of complex tasks in dynamic environments.
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Driving complex **project lifecycles**, ensuring seamless execution of design, testing, validation, and final **system deployment** to meet industrial standards.
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Pioneering innovation in **automation & manufacturing**, consistently enhancing productivity, accuracy, & **product quality** through state-of-the-art **machine vision technology**.

WORK EXPERIENCE

Dec' 24 - Sept' 25



Computer Vision Engineer (Video Analytics) | Techmahindra, Delhi

KeyResponsibilities:



Designed and implemented a real-time object detection and tracking system using the YOLO architecture, tailored for dynamic environments with live video feeds.



Integrated multi-object tracking algorithms to maintain persistent object identities across frames, ensuring accurate monitoring in real-time conditions.



Optimized model inference and video processing pipeline for low-latency performance, enabling reliable object detection and tracking in live deployment scenarios.



Developed and deployed an action recognition system for a healthcare compliance monitoring application, focusing on critical actions such as hand sanitization and contact with sensitive medical equipment.



Fine-tuned a SlowFast architecture (by Facebook AI) on the custom dataset for temporal action recognition, achieving 80% accuracy in real-world hospital environments on Azure ML platform.



Preprocessed video dataset to augment the data to train efficiently.

Apr'22- May 24



Trimble/Wabtec Corporation, Boston | Machine Vision Engineer

Key Responsibilities:



Engineered a deep learning algorithm utilizing the RCNN model achieving a 95% detecti operations for image/object detection tasks.



Conducted in-depth statistical analysis of existing systems, identifying bottlenecks and implementing targeted solutions to optimizing overall performance and productivity 30% improvement in overall system efficiency



Drove substantial revenue growth by 70% by ensuring timely product delivery and installation real-time issues, fostering effective cross-functional collaboration.



Utilized Debugging Programs to optimize image processing algorithms and enhance machine vision capabilities.

Feb'17- Oct'19



Assistant System Engineer| Tata Consultancy Services Ltd., Mumbai

Key Responsibilities:



Engineered advanced image processing algorithm, enhancing text extraction accuracy from



scanned documents and optimizing deployment via DevOps pipeline.



Led cross-functional team collaboration, streamlining deployment processes and driving substantial improvements in operational efficiency.



Applied expertise in writing and Debugging Programs to streamline the product development and deployment cycle.

INTERNSHIP

Robotics Software Intern, Southie Autonomy, Boston

Key Responsibilities:



Developed advanced object detection algorithms for robotic arms, leveraging multi-sensor data fusion (camera and LiDAR) and achieving a 40% improvement in detection accuracy for pick-and-place tasks in real-time.



Created robust template matching and feature extraction pipelines for robotic perception, improving system efficiency by reducing operational time by 30% through targeted bottleneck elimination.

RESEARCH WORK

Robotics Graduate Student Researcher, Worcester Polytechnic Institute

Key Responsibilities:



Developed and trained an auto-encoder based CNN model using PyTorch achieving 85% accuracy in estimating joint angles from the Kuka robot's pose as an innovative method to estimate joint angles using Artificial Intelligence.



Engineered an effective path planning algorithm using C++ and ROS, enabling autonomous movement while capturing depth images and increasing efficiency in operations thus achieving substantial accuracy in joint angle estimation from robot poses, enhancing prediction reliability.